

Eugene Lafayette Buchanan

Staff AI Test Automation Engineer — Agentic Automation Systems, SDET Lead

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github.com/eugeneb50/eugeneb50 — ZeroClaw Labs open-source agentic gateway (32.8k stars)

Professional Summary

Staff-level test automation engineer with 25+ years in Software Development Engineer in Test (SDET) and Quality Assurance (QA) practice, including Test Automation Lead and Product Owner roles. I build automation systems, not just test cases: end-to-end regression suites with health dashboards and alerting, CI/CD (continuous integration / continuous delivery) quality pipelines with gates on every change, and API-layer coverage at service boundaries. I evaluate AI-generated code in production, shipped a retrieval-augmented generation (RAG) support chatbot with guardrails, and contribute agentic AI infrastructure to a 32.8k-star open-source Rust gateway (agent lifecycle observability, model routing and cost attribution). Modern end-to-end frameworks are Cypress, Selenium, JUnit, and Postman; I ramp that page-object, fixture, and feedback-loop discipline to Playwright conventions. Git-based CI practice (Git, Bitbucket) transfers to GitLab pipelines, merge request workflows, parallelization, and caching.

Core Skills — Mapped to Gravie Requirements

AI-driven automation systems (core of this role): agentic AI infrastructure in production Rust gateway; evaluation of AI-generated code; RAG chatbot with context retrieval and guardrails; prompt, context-structure, and validation-pattern iteration through maintainer review; feedback loops between local runs, CI results, and resubmission

End-to-end automation (Playwright-transferable): Cypress, Selenium, JUnit, Postman, Qase; page-object and fixture discipline; end-to-end regression suites with health dashboards and alert systems; pass/fail trend, flakiness, and coverage-gap instrumentation used to aim new tests at highest-value gaps; white-box testing

API and contract-adjacent testing: REST API testing with Postman and custom APIs; webhooks; SFTP; service-boundary judgment (UI vs. API layer); integration verification across SAP, Oracle, Workday, Coursera, UKG, Zoom via SAML, OAuth, SFTP, SCORM, LTI (no Pact claim; no claims-system claim)

CI/CD and delivery: CI/CD quality pipelines (Apache, REST, SQL, AWS, React frontend); nightly build test script automation; quality gates per change; Git and Bitbucket merge workflows transferable to GitLab; parallelization and caching for fast feedback; Elasticsearch, S3, Docker

Migration practice: software evaluation and migration (accounting and medical-records software), nightly-build automation across OS generations, and large-suite maintenance with coverage continuity — transferable to Geb/Spock-to-Playwright conversion with AI agents (no Geb/Spock/Groovy claim)

Languages and stack: JavaScript, TypeScript, Python, Rust, PHP, Node.js, React, SQL; AWS (Elastic Compute Cloud [ECS pattern], Simple Storage Service [S3]); Apache; Git; Bitbucket; Elasticsearch; Scorm Cloud

Leadership and communication: hired and trained test automation team; Product Owner backlog with KPI tracking and Product Development Review (PDR) life cycles; cross-functional work with QA, Engineering, and Product; plain-language trade-off explanations, documentation, and SOP guides

Quality practice: SOC 2 compliance research and implementation; Blue Team incident forensics; regression, performance, load, security, usability, accessibility, and localization testing; technical debt reduction

Work Experience

Senior QA Engineer / Test Automation Lead / Product Owner — Integrations

Knowledgecity LLC, Remote, Dec 2020 – Aug 2025

- Built the automation system, not just test cases: hired and trained the test automation team and shipped end-to-end regression suites with health dashboards and alert systems covering complex deployment scenarios — the same pass/fail, flakiness, and coverage-gap instrumentation Gravie uses to aim AI test generation.
- Ran a CI/CD quality pipeline (Apache, REST, SQL, AWS, React) with Cypress, Selenium, Postman, JUnit, Elasticsearch, and S3 under Git/Bitbucket merge workflows — pipeline configuration, test stages, and

per-change quality gates transferable to GitLab, with parallelization and caching for fast feedback.

- Practiced API-vs-UI judgment daily: verified integrations at service boundaries (custom REST APIs, webhooks, SFTP, SAML, OAuth, SCORM, LTI) across SAP, Oracle, Workday, Coursera, UKG, and Zoom, reserving end-to-end UI tests for real user journeys.
- Iterated prompts, context structures, and validation patterns on a shipped AI support chatbot with RAG context retrieval and guardrails; evaluated AI-generated code under SOC 2 controls — directly transferable to raising AI first-pass test-generation rates.
- Led quality across integration, frontend, backend, mobile, security, performance, load, API, database, functionality, usability, accessibility, and localization testing; reduced technical debt and joined Blue Team incident forensics.
- Managed dev and test teams, product backlog, and PDR life cycles with KPI tracking as Integrations Product Owner; coordinated QA, Engineering, and Product on coverage, delivery timelines, and defect risk.

Open-Source Contributor — Agentic AI Infrastructure (hands-on agent workflows at scale)

ZeroClaw Labs, github.com/zeroclaw-labs/zeroclaw, 32.8k GitHub stars, 2026 – Present

- PR #7946 (merged Jul 2026, 39 commits): model context-window meter across TUI, gateway agent chat, and CLI covering 9 providers — one source of truth for model routing and cost attribution, the same provider/model bookkeeping AI test-generation infrastructure needs.
- PR #8966 (open, needs-maintainer-review, 80 commits): live serving-provider identity on usage events so usage, cost, and context attribution follow the provider that actually served the call across fallback and model switches, with per-attempt ledger and failure-recovery policy.
- PR #8337 (closed, superseded by #10269): agent lifecycle observer (idle/working/blocked/released) over JSON-RPC on Unix domain sockets — feedback-loop infrastructure letting agents run, interpret results, and resubmit; cleared four rounds of maintainer review and its requirements were adopted into the vendor-neutral lifecycle design.
- Daily hands-on agentic development: build-validate-fix-resubmit loops with AI coding agents, structured context assembly, and reviewer-driven validation — the exact workflow Gravi applies to AI-generated Playwright tests.

Senior Software Engineer II (SDET Practice)

RealNetworks, Seattle, WA, 1999 – 2001

- SDET quality assurance across Linux, Unix, Windows, and embedded systems for streaming media platform, servers, and mobile/consumer appliances, including TFRCP patent work.
- Global presales engineering with vice president sales support, including travel to Japan, Korea, and the continental United States; trained new hires.

Earlier Software Test Engineering and Support Roles

Microsoft, IBM, Keene Inc., 1996 – 1999

- Microsoft, Redmond, WA (1999): Senior Software Test Engineer IV, Windows Media Server versions 4 and 5; nightly build test script automation — early local-execution-to-CI feedback loop.
- IBM, Kirkland, WA (1998): Senior Test Engineer III for high-availability servers; WHQL certification for cluster failover.
- Microsoft, Redmond, WA (1997): Software Test Engineer II for Windows 98 and Windows NT; original equipment manufacturer (OEM) setup and hardware/driver verification.
- Keene Inc., Seattle, WA (1996): Technical Support Agent II for Windows 95 and Windows NT; frontline and escalated support ticketing.

Business Consultant — Campus IT, Software Evaluation and Migration

Medico Investments LLC / Foremost Senior Care / Foremost Organization, Hesperia, CA, 2013 – 2016

- Evaluated and migrated accounting and medical-records software — migration discipline (coverage continuity during conversion) transferable to legacy-to-modern framework migration.
- Ran campus IT support: security and backup policies for servers and PCs, video surveillance upgrade, campus-wide wireless; set up sound/video and ran events.

IT Manager

Lucerne Valley Unified School District, Lucerne Valley, CA, 2002

- Supported four campuses (Wide Area Network [WAN]/Local Area Network [LAN], NetWare, Windows, Linux, virtualized desktops, audio/video).
- Met state attendance submission requirements via data Extract, Transform, and Load (ETL) with table locking and certification.

Education

- Associate Degree, Victor Valley College
- California Notary Commission
- Toastmasters International (technical trade-off communication and documentation)